

IMANTS ZVIRGZDINS

MECHANICAL ENGINEER
INDUSTRIAL DESIGNER

10+ YEARS EXPERIENCE

RIGA, LATVIA

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MICROPHONE HANDLING-NOISE REDUCTION / BUBBLEBEE INDUSTRIES

Product engineering with industrial-design range.

Mechanical engineer and industrial designer with 10+ years of experience developing physical products from early architecture through prototyping, validation, production documentation, and supplier handoff.

CORE CAPABILITIES

Mechanical product design

Mechanisms, enclosures, electronics packaging, tolerance-aware details, and assembly definition.

CAD and documentation

SolidWorks, controlled assemblies, production drawings, BOMs, manuals, and PDM workflows.

Manufacturing and DFM

Injection moulding, CNC, sheet metal, extrusion, additive manufacturing, and supplier communication.

Validation

FEA/CFD support, vibration and transmittance testing, ESP32 data logging, and assembly-process test jigs.

PROFESSIONAL EXPERIENCE

■	2025.03 - 2026.07	Senior Mechanical Engineer Bubblebee Industries Audio hardware, quick-release mechanisms, damping layouts, and ESP32 acceleration validation.
■	2023.10 - 2025.02	Senior Mechanical Engineer Giraffe360 LiDAR motor R&D, optical tooling, injection-moulded parts, CFD, jigs, and production support.
■	2023.06 - 2023.08	Mechanical Designer Warp Agency Mechanical architecture and prototypes for a quiet 2 x 7 m kinetic robotic chandelier.
■	2022.09 - 2023.10	Mechanical Engineer / Industrial Designer Riga Technical University Sealed sensors, industrial mechanisms, startup prototypes, calculations, and technical layouts.
■	2015.09 - 2022.09	Lead Mechanical Engineer / Industrial Designer SAFtehnika JSC RF equipment and IoT products from concept through mass production, IP54/IP68, and ATEX support.
■	2013 - 2015.09	Industrial Designer / Project Manager ET Sons Complex CNC-manufactured design objects and wearable product concepts.

BUBBLEBEE INDUSTRIES / 2025.03 - 2026.07

Microphone handling-noise reduction



Professional audio hardware focused on reducing transmitted handling noise while keeping microphone setup fast and production-ready.

ENGINEERING CONTRIBUTION

- Designed mechanical quick-release concepts and blimp hardware.
- Developed damping layouts with strong emphasis on material choice and efficient assembly flow.
- Compared mechanical configurations through vibration and transmittance analysis.
- Built an ESP32-based acceleration test jig with custom data logging for validation.

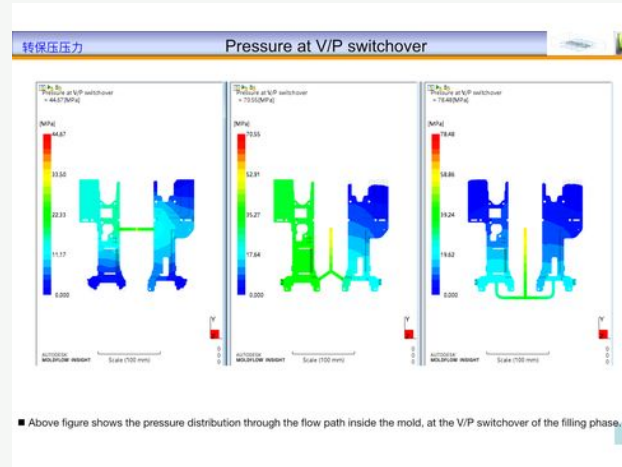
PROCESS

Concept architecture / CAD / prototype build / vibration measurement / design iteration / production detail



GIRAFFE360 / 2023.10 - 2025.02

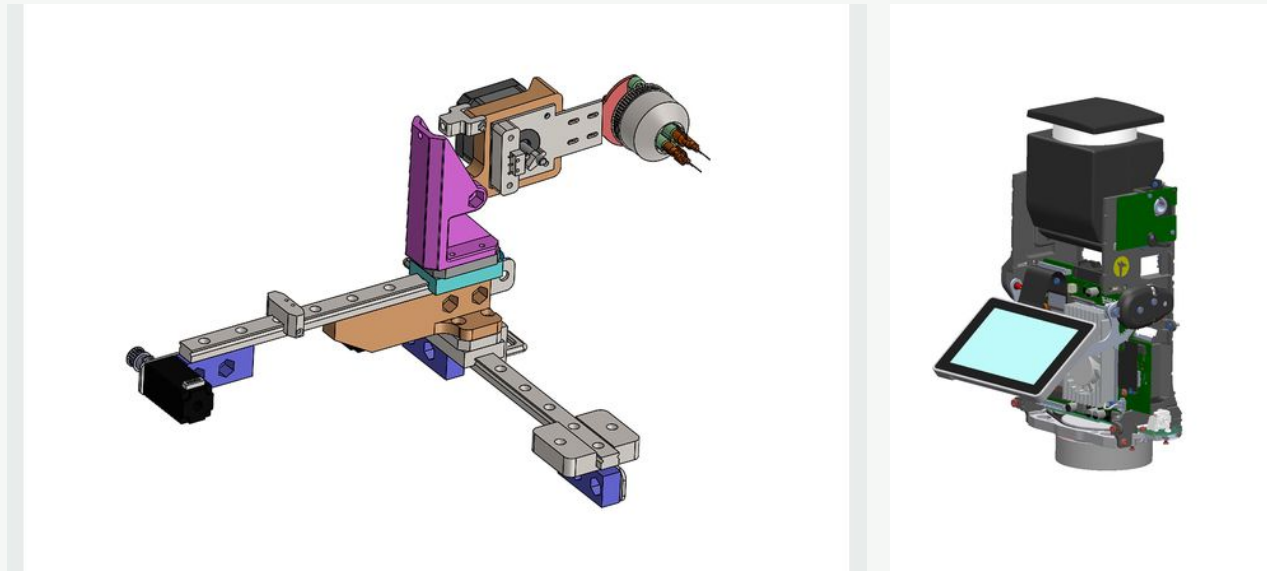
LiDAR motor and optical tooling



Mechanical R&D and production support for a LiDAR-integrated camera platform, connecting rotating hardware, optical alignment, electronics packaging, and assembly process design.

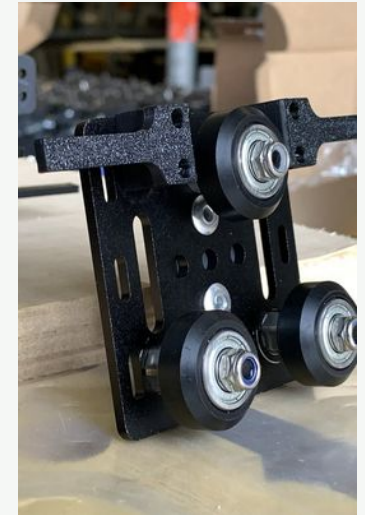
SELECTED RESPONSIBILITIES

- LiDAR-integrated brushless motor development and injection-moulded part design.
- Optical component gluing robot, adjustment frames, and PCB test jigs.
- Moldflow pressure and deflection analysis for the RJC side-bracket tooling.
- Assembly issue investigation and supplier communication for process reliability.



WARP AGENCY / GRAVITY TEAM LUSTRA / 2023.06 - 2023.08

2 x 7 m kinetic robotic chandelier



Mechanical architecture and workshop validation for a large-format chandelier designed to move quietly, reliably, and repeatedly above an occupied interior.

- Rail, pulley, actuator, support, and ceiling-interface design.
- Low-noise motion and robust repeated operation.
- Workshop prototypes before full ceiling installation.
- SolidWorks assemblies, part models, and installation-ready details.

SAFTEHNIKA JSC / LMT / 2015.09 - 2022.09

Spectrum Compact and drone integration



LEAD MECHANICAL ENGINEERING

Portable RF measurement hardware developed for field installers and regulatory institutions, from product architecture through production release.

SPECTRUM COMPACT

- Owned mechanical architecture, industrial design, and manufacturable product detail.
- Combined CNC, sheet metal, silicone moulding, PCB packaging, and IP54 requirements.
- Prepared production drawings, assembly documentation, BOMs, and supplier-ready files.

DRONE MODIFICATION WITH LMT

- Developed a lightweight modular attachment for RF tower monitoring and audit workflows.
- Integrated vibration damping, exchangeable RF polarization components, and infrared-camera support.



SAFTEHNIKA JSC / ARANET / 2015.09 - 2022.09

IoT sensors and 5G radio equipment



ARANET

Industrial design and mechanical engineering for LoRa sensor products used in farming, horticulture, offices, and education.

- Injection-moulded parts, IP68 solutions, drawings, BOMs, and quality control.
- Aranet4 exceeded 100,000 produced units and was Latvia's Most Innovative Product of 2019.

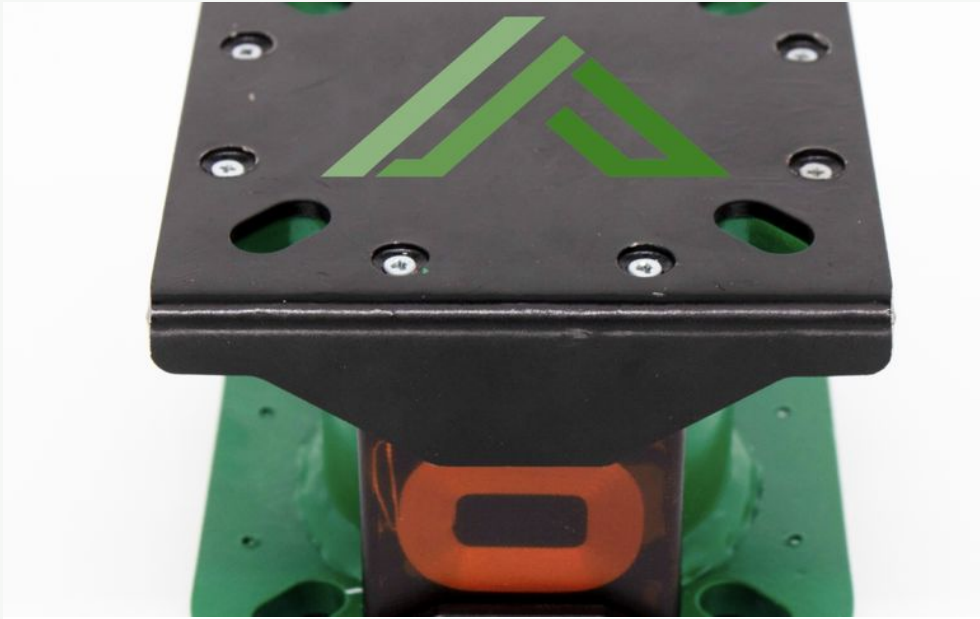
INTEGRA-E2

Lead mechanical engineering for point-to-point 5G radio equipment with a simple, robust enclosure philosophy.

- Aluminium-extrusion base and CNC-machined heat-sink cover.
- IP68 design, ATEX certification support, assembly definition, and BOM documentation.



Industrial sensors and process mechanisms



SCAFFOLDING PRESSURE SENSOR

Load-cell based sensor with sealed mechanical architecture and wireless operation.

- PU overmoulding, LoRa antenna, wireless charging, and IP68 protection.
- Sheet-metal covers, PCB layout, battery assembly, and FEA support.



CARBON SCRAPER MECHANISM

Mechanical scraper concept, layout, calculations, and documentation for an industrial carbon-processing mechanism.

- Scraper architecture and industrial process layout.
- Mechanical load calculations, FEA stress analysis, and production drawings.

SELECTED BREADTH / 2013 - 2023

Industrial design and production detail



FIRE ALARM SYSTEM

EN-54 product-family design for smoke, sound, light, and control devices.



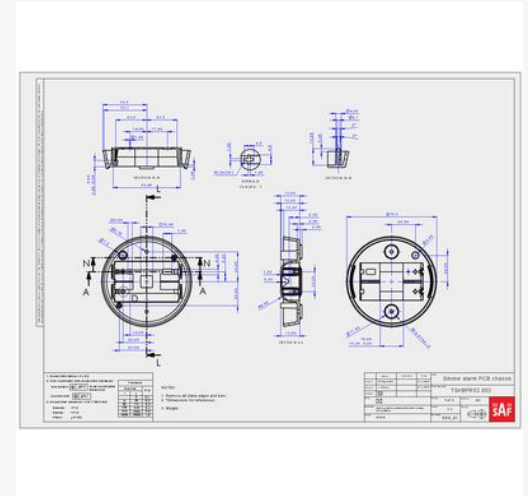
CHEMICAL GAS SENSOR

Compact IP68 concept with a simplified airtight, screw-free assembly.



LATVIAN ARCHITECTURE AWARD

50+ part Rubik's-cube pineapple produced with 5-axis CNC milling.



PRODUCTION DRAWINGS

Supplier-ready part, assembly, exploded-view, and quality-control documentation.

Certifications and training

SolidWorks certification and specialist training supporting mechanical design, simulation, plastics, and production-ready product development.



DASSAULT SYSTEMES SOLIDWORKS

SOLIDWORKS Professional - Mechanical Design

Academic exam at Riga Technical University / 2023.03.14



DASSAULT SYSTEMES SOLIDWORKS

SOLIDWORKS Associate - Mechanical Design

Academic exam at Riga Technical University / 2023.03.10



PLM GROUP LATVIJA SIA

SolidWorks Simulation Training

Specialist training in simulation workflows for mechanical product development.

2016.03.17-18



PLM GROUP LATVIJA SIA

SOLIDWORKS Plastics Training

Specialist training in plastics design and injection-moulding analysis workflows.

2016.11.10-11